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1-DESCRIPTION

Offset adjustments for ACGD Cabinet are automated using Digital DC Offset; however, some FEs continue to use the DVM approach that worked on SGD: measuring at the SGA output with a Digital Volt Meter (DVM). DVMs will give an inaccurate reading of offset voltage with the switching waveforms on the output of the SGA sometimes as high as 30Vdc when the actual offset is <10mV.

A procedure for monitoring DC offsets has been developed for the ACGD. This procedure is for troubleshooting only and is to be used to monitor DC Offsets. If Offsets are suspect for causing system malfunction, contact the OLC to get engineering support.

ACGD amplifiers operate in a switching state when enabled. Voltage **CANNOT** be measured using a DVM at the SGA output terminals.

Required Tools:

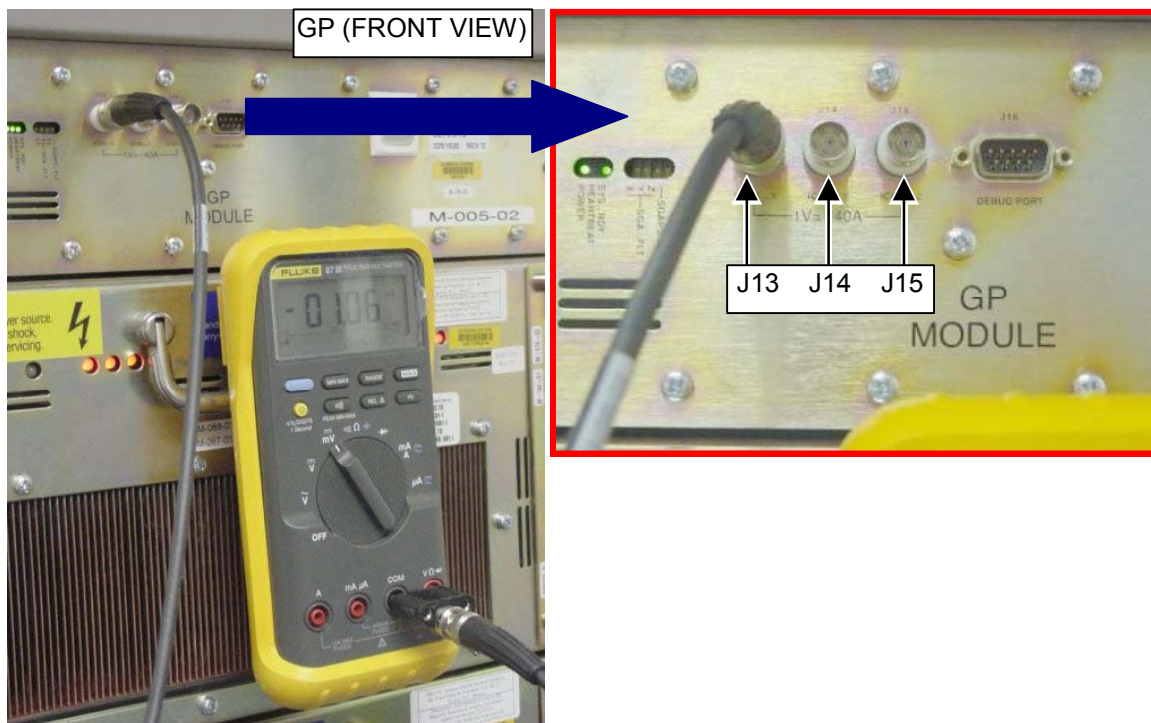
- Digital Volt Meter (DVM)
- BNC to Banana Plug adaptor for the DVM

2-MONITORING DC OFFSETS

Note

If unsure whether the ACGD is in a disabled state, wait 15 minutes without scanning and it will disable. Another way to disable the ACGD is to power cycle the GP. If the SGA is enabled, a large group of LEDs on the SGA Control board will be lit. When the ACGD is disabled, only a few LEDs will be lit on each of the SGAs.

1. With the ACGD cabinet powered on and in a disabled state, use a DVM (in the mV scale) to measure J13, J14, and J15 (X, Y, and Z axis respectively) on the front panel of the GP, see Illustration 2-1. Multiply these voltages by 40 to obtain reference currents, in mA, for each axis. Record measurement in Table 2-1.



MEASUREMENT LOCATIONS ON THE GP
ILLUSTRATION 2-1

Note

To enable ACGD, prescribe any scan until the Save Scan button is active. Do not start the scan. Gradshim values need to be set to Zero. The FE should wait 5 minutes after scanning or prescanning to measure offsets because the very-long-time-constant eddy currents will look like a DC offsets until they have settled.

2. With the ACGD cabinet powered on and enabled, use a DVM to measure again J13, J14, and J15. Multiply these voltages by 40 to obtain currents, and compare them to the "reference" currents measured in step 1.

2-MONITORING DC OFFSETS (Continued)

3. The *difference* between the reference current and the enabled current is the offset, refer to Table 2-1 to determine offset value. For example:
- ACGD Disabled, measure J13 to be 3 mV * 40 = 120 mA.
 - ACGD Enabled, measure J13 to be 2.8 mV * 40 = 112 mA.
 - DC Offset is 120 mA - 112 mA = 8 mA.
4. If it is necessary to adjust the offsets, use the Digital DC Offset diagnostic. **DO NOT ADJUST THE POTENTIOMETERS ON THE SGAs** (these are used **ONLY** for factory calibration). If the potentiometers are adjusted the module may change the tuning to a range that the DC Offsets would not be able to bring to zero, thus causing a hard failure.

TABLE 2-1
DC OFFSET MEASUREMENT

DC Offset Measurement			
Axis ↓:	Disabled State	Enabled State	Offset (Disabled – Enabled)
X (J13)	_____ mV *40 = _____ mA	_____ mV *40 = _____ mA	_____ mA
Y (J14)	_____ mV *40 = _____ mA	_____ mV *40 = _____ mA	_____ mA
Z (J15)	_____ mV *40 = _____ mA	_____ mV *40 = _____ mA	_____ mA

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	May 7, 2003	K. Keshena	Preliminary Version.
0	May 16, 2003	K. Keshena	Initial release.